NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

INCH-POUND

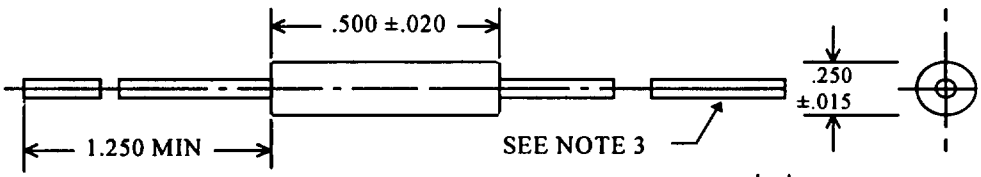
MIL-PRF-20/30B
9 May 1989
SUPERSEDING
MIL-C-20/30A
26 August 1976

PERFORMANCE SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (TEMPERATURE COMPENSATING), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, STYLES CCR78 AND CC78

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-20.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Lead diameter shall be .015 (0.38 mm) to .027 (0.69 mm).
Ⓑ 4. Lead length may be a minimum of 1.00 inch (25.4 mm) long for use in tape and reel packaging, when specified in the ordering data.

Inches	mm
.015	0.38
.020	0.51
.027	0.69
.250	6.35
.500	12.70
1.250	31.75

FIGURE 1. Styles CCR78 and CC78 capacitors.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Lead type: Axial.

Case type: Tubular, molded.

DC rated voltage: See table I.

Operating temperature range: -55°C to +125°C.

Characteristic: CG.

Ⓑ Failure rate level (CCR78 only): M (1.0%), P (0.1%), R (.01%) or S (.001%).

Ⓑ denotes changes

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Temperature cycling and voltage conditioning (CCR78 only): In accordance with MIL-C-20.

Capacitance: Within tolerance specified (see table I).

Dissipation factor: In accordance with MIL-C-20.

Dielectric withstanding voltage: In accordance with MIL-C-20.

Body insulation: Test II.

Solderability: In accordance with MIL-C-20.

Resistance to soldering heat: In accordance with MIL-C-20.

Life: In accordance with MIL-C-20, operating condition 2.

Marking: In accordance with MIL-C-20, method I.

TABLE I. Capacitor characteristics.

Type 1/ designation	Rated voltage (volts, dc)	Nom. cap. (pF)	Cap. tol.	Type 1/ designation	Rated voltage (volts, dc)	Nom. cap. (pF)	Cap. tol.
CC-78CG821--	200	820	FGJK	CC-78CG562--	100	5,600	FGJK
CC-78CG102--	200	1,000	FGJK	CC-78CG682--	100	6,800	FGJK
CC-78CG122--	200	1,200	FGJK	CC-78CG822--	100	8,200	FGJK
CC-78CG152--	200	1,500	FGJK	CC-78CG103--	100	10,000	FGJK
CC-78CG182--	200	1,800	FGJK	CC-78CG123--	100	12,000	FGJK
CC-78CG222--	200	2,200	FGJK	CC-78CG153--	50	15,000	FGJK
CC-78CG272--	200	2,700	FGJK	CC-78CG183--	50	18,000	FGJK
CC-78CG332--	200	3,300	FGJK	CC-78CG223--	50	22,000	FGJK
CC-78CG392--	100	3,900	FGJK	CC-78CG273--	50	27,000	FGJK
CC-78CG472--	100	4,700	FGJK				

1/ Complete type designation will include the following:

- 1st dash - Symbol "R" (for style CCR78) or dash will be deleted (for style CC78).
  2nd dash - Applicable capacitance tolerance symbol.
  3rd dash - Applicable failure rate level symbol (CCR78 only) or dash will be deleted (for style CC78).

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Navy - OS
Air Force - 99
DLA - ES

User activity:
Navy - AS, MC, SH

Preparing activity:
Air Force - 85

Agent:
DLA - ES

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